Set partition

- Stirling numbers of the second kind
  (the Stirling numbers of the first kind enumerate graphs with cycles, will be introduced later)

We have a set of distinguishable elements (numbers). How many partitions we might have to k subsets?

e.g. $S(5,2) = 15 = \left\{ {5 \atop 2} \right\}$ $\qquad \boxed{S(n,k) = \left\{ {n \atop k} \right\}}$

$\{1\},\{2,3,4,5\}$,	$\{1,2\}\ldots$	$\{2,3\}\ldots$	$\{3,4\}\ldots$
$\{2\},\ldots$	$\{1,3\}\ldots$	$\{2,4\}\ldots$	$\{3,5\}\ldots$
$\{3\}\ldots$	$\{1,4\}\ldots$	$\{2,5\}\ldots$	$\{4,5\}\ldots$
$\{4\}\ldots$	$\{1,5\}\ldots$		
$\{5\},\ldots$			

If we already have the value $S(n-1, k)$ - number of partitions of n elements to k subset, when we add another element, we can have k elements by creating singleton apart from $(k-1)$ sets or add to one of k datasets, so

$$\boxed{S(n,k) = S(n-1, k-1) + k\,S(n-1, k)}$$

see also, that $S(0,0) = 1$, $S(0,k) = 0$ for $k > 1$

also $S(n,1) = S(n,n) = 1$

so

$$S(5,2) = S(4,1) + 2 \cdot S(4,2)$$

$$S(4,1) = 1$$

$$S(4,2) = S(3,1) + 2 \cdot S(3,2) = 7$$

because $S(3,2) = \binom{3}{1}$ as $\{1,2,3\} \to \{1\}, \{2,3\}$
$\{2\}, \{1,3\}$
$\{3\}, \{1,2\}$

so $S(5,2) = 1 + 2 \cdot 7 = 15$

Indeed.

Exercises:

1) Calculate

a) $S(5,3) = 25$

b) $S(7,5) = 140$

c) $S(n,n-1) = \binom{n}{2} = \frac{n(n-1)}{2}$ ← one subset has to be 2-element

d) $S(n,2) = 2^{n-1} - 1$

we have 2 non-empty sets.

so we have two states for each element → 2^n, but we have to adjust for swapping the sets and remove empty and full set

so $\frac{2^n - 2}{2} = 2^{n-1} - 1$

2) List all partitions of 7-element set to 2 blocks:

$1 \mid 2,3,4,5,6,7$ $\quad$ $1,2 \mid 3,4,5,6,7$ $\quad$ $1,2,3 \mid 4,5,6,7$ $\quad$ =

⋮

$$= \binom{7}{1} + \binom{7}{2} + \binom{7}{3} = 7 + 21 + 35 = 63 = 2^6 - 1 = 2^{7-1} - 1$$

3. List all partitions of 5-element set:

1 set $1,2,3,4,5$ $\qquad S(5,1) = 1$

1 option

2 sets $\quad 5 = 1+4 \quad 1 \mid 2,3,4,5 \quad \binom{5}{1} = 5$

$5 = 2+3 \quad 1,2 \mid 3,4,5 \quad \binom{5}{2} = 10$

$S(5,2) = 15$

3 sets $\quad 5 = 1+1+3 \quad 1 \mid 2 \mid 3,4,5 \quad \binom{5}{3} = 10$

$5 = 1+2+2 \quad 1 \mid 2,3 \mid 4,5 \quad \frac{S\binom{4}{2}}{2} = 15$

$S(5,3) = 25$

4 sets $\quad 5 = 1+1+1+2 \quad 1 \mid 2 \mid 3 \mid 4,5 \quad \binom{5}{2} = 10$

$S(5,4) = 10$

5 sets $\quad S(5,5) = 1$

$$\sum_{k=1}^{5} S(5,k) = 1 + 15 + 25 + 10 + 1 = 52 = B_5$$

$\boxed{\text{The sum } \sum_{k=1}^{n} S(n,k) \to \text{the total number of partitions is called the Bell number } B_n}$

4 Find B_7

$$B_7 = S(7,1) + S(7,2) + S(7,3) + S(7,4) + S(7,5) + S(7,6) + S(7,7)$$

$S(7,1) = 1$

$S(7,2) = 63$

$S(7,3) = 301$

$S(7,4) = 350$

$S(7,5) = 140$

$S(7,6) = 21$

$S(7,7) = 1$

$B_7 = 877$

Table for $S(n,k)$

n\k	1	2	3	4	5	6	7
1	1						
2	1	1					
3	1	3	1				
4	1	7	6	1			
5	1	15	25	10	1		
6	1	31	90	65	15	1	
7	1	63	301	350	140	21	1

We will show

$$S(n,k) = \frac{1}{k!} \sum_{i=0}^{k} (-1)^i \binom{k}{i} (k-i)^n$$

For Bell numbers it's easier

$$B_n = \sum_{k=1}^{n} S(n,k)$$

$$B_{n+1} = \sum_{k=1}^{n} \binom{n}{k} B_k$$

Proof: suppose we have $\{1, \dots, n, n+1\}$

Take $n+1$. The block containing $(n+1)$ will have k other elements. So we might choose them in $\binom{n}{k}$ ways

remaining $n-k$ elements have their own partition $\rightarrow B_{n-k}$

so $B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_{n-k}$

reindex $i = n-k \rightarrow \binom{n}{k} = \binom{n}{i}$, the sum does not change

so

$$\boxed{B_{n+1} = \sum_{k=0}^{n} \binom{n}{i} B_i}$$

Maps

How many maps there are from $N \to R$?

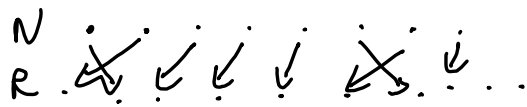
$|N| = n$, $|R| = r$

→ In general each element $x \in N$ could be mapped to any $y \in R$ so there are r "states" for each $x \in N$, so there are

r^n possible maps.

→ How many maps are injective? (1-1)

For each $x \in N$ we select exactly one $y \in R$

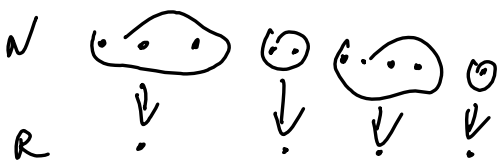


so we have $r(r-1)(r-2)\dots(r-n)$ ways to do so, which is equal

$$\boxed{\frac{r!}{(r-n)!} = \binom{r}{n}(r-n)!}$$

→ How many are surjective?

We need $f: N \to R$ such, that $\forall y \in R$ $f^{-1}(y) \in N$, so



we have to partition the set N into k sets $S(n,r)$ but there are $r!$ ways to do assignment so →

Number of surjective maps is

$$S(n,r)\cdot r!$$

Note, that not all maps are surjective, as in some cases some elements are not in the image. But there is always a subset of k elements when the map is surjective. So we get:

$$r^n = \sum_{k=1}^{r} \binom{r}{k} \cdot S(n,r)\, r!$$

Partitions of identical objects

We analyzed cases

$5 = 1+4$, $5 = 2+3$ etc.

How many partitions of number n as a sum of k elements with each non-zero?

Lets call this number $p(n,k)$

For example

$p(5,2) = 2$ as there are $5=2+3$, $5=1+4$ or

$\begin{smallmatrix} \bullet & \bullet & \bullet & \bullet \\ \bullet & & & \end{smallmatrix}$ or $\begin{smallmatrix} \bullet & \bullet & \bullet \\ \bullet & \bullet & \end{smallmatrix}$ $\leftarrow$ $\boxed{\text{these diagrams are called Ferret diagrams}}$

ordered from longest to shortest!

Observe that

$p(n,k)$ is the number of Ferret diagrams with n dots and k rows

There are two exclusive cases:

1) either the shortest row has 1 dot
2) the shortest row has > 1 dot

In 1) there are exactly $p(n-1, k-1)$ such diagrams (from definition)

In 2) we have "full" column on the left and removing it leaves us with

$p(n-k, k)$ such graphs. So

$$p(n,k) = p(n-1, k-1) + p(n-k, k)$$

To my knowledge there is no closed formula for $p(n,k)$

Still observe

$p(n,1) = 1$ we take $p(0,0) = 1$

$p(n,n) = 1$ $p(0,n) = 0$ for $n > 0$

Also

$p(n, n-1) = 1$ (one "2")

$p(n, n-2) = 2$ (either one "3" or two "2")

Exercises

1) Calculate

a) $p(9,5) = 5$

b) $p(11,4) = 11$

c) $p(13,8) = 7$

d) $p(n,2),\ n \geq 2 = \lfloor n/2 \rfloor$

a) $9 = 1+1+1+1+5$

$9 = 1+1+1+2+4$

$9 = 1+1+1+3+3$

$9 = 1+1+2+2+3$

$9 = 1+2+2+2+2$

b) $p(11,4) = p(10,3) + p(7,4) = 8+3 = 11$

$p(10,3) = p(9,2) + p(7,3) = 4+3 = 8$

$p(9,2) = \underset{\substack{\shortparallel \\ 1}}{p(8,1)} + \underset{\substack{\shortparallel \\ 3}}{p(7,2)} = 4$

$p(7,3) = \underset{\substack{\shortparallel \\ 3}}{p(6,2)} + \underset{\substack{\shortparallel \\ 1}}{p(4,3)} = 4$

$p(7,2) = \underset{\substack{\shortparallel \\ 1}}{p(6,1)} + \underset{\substack{\shortparallel \\ 2}}{p(5,2)} = 3$

$p(7,4) = \underset{\substack{\shortparallel \\ 3}}{p(6,3)} + \underset{\substack{\shortparallel \\ 0}}{p(3,4)} = 3$

$p(6,3) = \underset{\substack{\shortparallel \\ 2}}{p(5,2)} + \underset{\substack{\shortparallel \\ 1}}{p(3,3)} = 3$

Table

$n \backslash k$	1	2	3	4	5	6	7	8	9	10
1	1									
2	1	1								
3	1	1	1							
4	1	2	1	1						
5	1	2	2	1	1					
6	1	3	3	2	1	1				
7	1	3	4	3	2	1	1			
8	1	4	5	5	3	2	1	1		
9	1	4	7	6	5	3	2	1	1	
10	1	5	8	9	7	5	3	2	1	1

Observe also, that

$$p(n,k) = \sum_{i=1}^{k} p(n-k,i) = p(n-k,1) + p(n-k,2) + p(n-k,3) + \dots + p(n-k,k)$$

if $n = a_1 + a_2 + \dots + a_k$ with $a_i \geq 1$

then $(a_1 - 1) + (a_2 - 1) + \dots + (a_k - 1) = n-k$

and some a_i values might be 0.

so this partition is in fact partition of $n-k$ element to at most k parts

And there is an obvious bijection

In terms of Ferrers diagrams

Partition of n to k parts is a diagram with k rows and n dots in total.

But this is equal to partition in $n-k$ dots to k rows at most (some may be empty), as we will "add" the left column.

So

$$p(7,3) = p(4,1) + p(4,2) + p(4,3) = 1+2+1=4$$

Also

$$p(13,8) = p(5,1) + p(5,2) + p(5,3) + p(5,4) + p(5,5) =$$

$$= 1 + 2 + 2 + 1 + 1 = 7$$

In fact if $n-k > k$ it is much easier to use this method.

For example

$$p(100, 97) = p(3) = p(3,1) + p(3,2) + p(3,3) = 1+1+1=3$$